

Cofunction identities



- 1 a 56° b 28° c $\frac{\pi}{3}$ d $\frac{3\pi}{14}$
- 2 a $\sin 53^\circ$ b $\cos 14^\circ$ c $\cot 22^\circ$ d $\tan 47^\circ$
e $\sec 55^\circ$ f $\sin \frac{7\pi}{18}$ g $\cos \frac{2\pi}{5}$ h $\tan \frac{7\pi}{18}$
i $\csc \frac{7\pi}{18}$ j $\sin \left(\frac{5\pi - 2}{10} \right)$
- 3 a i $\frac{a}{c}$ ii $\frac{a}{c}$ iii $\frac{b}{c}$ iv $\frac{b}{c}$
b If two angles are complementary then the sine of one angle will equal the cosine of the other angle.
c 0.32
d 90°
- 4 a $\cos p$ b $\tan x$ c $\frac{1}{7}$ d 1
e $\sin y$
- 5 0.19
- 6 0.46
- 7 a No solution provided. b No solution provided.

Trigonometric equations

- 8 a $\theta = 65^\circ$ b $\theta = 5^\circ$ c $\theta = \frac{3\pi}{8}$ d $\theta = \frac{\pi}{10}$

9 $c = \frac{5\pi}{36}$



10 $c = 5$

11 $B = 14$

12 $A = 4896$

13 a $x = 5^\circ$

b $x = \frac{\pi}{180}$